

Financial Education and Digital Asset Management: What's in the Black Box?

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Highlights

- The provision of financial education increases the tendency to delegate financial decision-making.
- Subjects receiving financial education display lower self-assessed risk aversion while their financial decision-making suggests higher risk aversion.
- Financial education enhances diversification while increasing decision delegation.

Financial Education and Digital Asset Management: What's in the Black Box?*

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Abstract

We study the effect of a financial education intervention on the propensity to delegate financial decision-making to a digital asset management tool (robo-advisor). Our results suggest that investors are more inclined to delegate financial decision-making when they receive detailed information on the investment principles underlying the robo-advisory algorithm. The intervention also affects investors' independent investment behavior as it increases diversification and revealed risk aversion, while it decreases self-reported risk aversion. Our findings imply that robo-advisors benefit from providing financial education to prospective investors and that financial education may indirectly affect investment outcomes by encouraging the use of financial advice.

JEL classification: D14, G11, G41, O33

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1. Introduction

Retail investors tend to make detrimental investment decisions, which are associated with substantial costs: Calvet et al. (2007) estimate that under-diversification alone caused the median Swedish household an annual return loss of 2.9 percentage points. Measures to improve financial decision-making usually involve either (the regulation of) professional financial advice or the provision of large-scale financial education programs (OECD, 2017). Findings from the financial advice literature suggest that the two measures may interact. Thus, using a randomized control trial, we investigate the potential interaction of financial advice and financial education. Specifically, we study the effect of a financial education intervention on the use of digital asset managers (robo-advisors), who can provide financial advice to less wealthy and less financially literate investors. Besides, robo-advisors constitute an interesting case study since there are calls for mandatory disclosure requirements regarding the applied algorithms. The requirements aim to communicate the underlying decision principles, which would be conceptually similar to a financial education intervention.

Prior literature has established that an investor's level of financial literacy is related to the tendency to seek and follow financial advice (e.g., Calcagno and Monticone, 2015). However, financial literacy is typically measured using a questionnaire and not induced through an education intervention, which makes identifying a causal relationship challenging. On the other hand, studies of financial education typically do not take into account the use of financial advice, which may lead to biased estimates of the effect on financial outcomes. In addition, while investors' interaction with algorithmic advice likely differs from the interaction with a traditional human advisor (Dietvorst et al., 2015), few studies in the financial advice context have specifically investigated the interaction with robo-advisors (e.g., D'Acunto et al., 2019). Finally, the effectiveness of financial education interventions is mainly assessed using changes in elicited financial literacy, reported financial behavior, or aggregate savings measures such as pension plan contributions (cf. Stolper and Walter, 2017). Portfolio allocation and maintenance decisions, though highly consequential, are rarely addressed.

This study contributes to the existing literature on financial education by including the use of financial advice in assessing the effectiveness of an intervention. Specifically, study participants were able to delegate parts of their experimental budget to a rule-based, algorithmic

asset management tool. In doing so, our study further sheds light on how transparency affects the acceptance of algorithmic decision-making in a sensible application area. At the same time, we add to existing literature on the use of financial advice by addressing endogeneity issues of studies using measured financial literacy by conducting a randomized control trial in a laboratory environment (cf. Fernandes et al., 2014). Finally, we focus our analysis on portfolio allocation choices, which thus far has been an understudied outcome dimension of financial education.

Our results suggest that a financial education intervention increases the extent to which investors delegate to a robo-advisory tool. The effect is stable across multiple specifications and is highly significant in both the statistical and economic sense. The budget share delegated by treated participants exceeds the share delegated by participants in the control group by more than 50 percent. This finding suggests that robo-advisors benefit from opening the “black box” of algorithmic financial decision-making to their prospective investors. It further implies that financial education can indirectly affect investment outcomes by encouraging the use of financial advice.

Aside from its effect on the tendency to delegate to the robo-advisor, the financial education intervention had spillover effects on the investors’ independent investment decisions. First, the intervention improved diversification, which did not displace delegation. Second, the intervention decreased self-reported risk aversion, while it increased the level of risk aversion as inferred from observed investment behavior in the laboratory. These findings imply that interventions that aim to increase the understanding of a robo-advisor’s decision-making process affect investor behavior beyond the delegation decision.

2. Background and related literature

Retail investor behavior and financial literacy

Retail investors tend to be subject to several costly investment mistakes. For instance, few investors are invested in the stock market despite the evidence of a substantial equity premium (Guiso et al., 2002). If they are invested, they are often under-diversified, exhibit an excessive preference for local equity (Goetzmann and Kumar, 2008), or insufficiently adjust their portfolios to general market movements (Calvet et al., 2009). Overconfident investors tend to trade excessively, which compromises their returns through transaction costs (Barber and

Odean, 2000; Glaser and Weber, 2007). Investors further tend to prefer to realize stock price gains rather than losses, thereby diluting investment performance (the so-called *disposition effect*, Shefrin and Statman, 1985). Surveys have shown that a large share of retail investors lacks understanding of basic financial concepts (Inderst and Ottaviani, 2012; Lusardi and Mitchell, 2011), which is associated with inefficient financial behavior (Agarwal et al., 2017). Although the established relationship is likely endogenous and thus overstated in magnitude (Hastings et al., 2013; Skagerlund et al., 2018), financial education programs aiming to increase financial literacy and improve financial decision-making have been introduced in a number of countries.

Financial education

Scholars have drawn on a large number of (mostly publicly funded) programs to estimate the effects of financial education on measures of financial behavior such as financial product awareness, contribution to savings plans, and credit card balances (cf. Hastings et al., 2013). However, establishing a causal relationship is tricky since participation is often voluntary and thus subject to selection. For instance, Meier and Sprenger (2013) report that participants of financial education programs are significantly more patient than non-participants. As a consequence, studies not accounting for selection bias overestimate the true effect of financial education interventions. Besides, Fernandes et al. (2014) conduct an extensive meta-analysis of 201 studies assessing the relationship between (measured and manipulated) financial literacy on various measures of financial behavior. Their findings suggest that the relationship between financial literacy and financial behavior is subject to omitted variable bias as the correlation is significantly lower in studies using manipulated financial literacy and studies employing an instrumental variable approach than in studies based on simple correlations. They also find that the average effect of financial education interventions on financial behavior is negligible and decays over time. In light of the disparity between costs and benefits of financial education programs, Willis (2011) suggests relying on professional financial advisors in achieving favorable investment outcomes. In her view, well-regulated financial advice poses a superior alternative to large-scale financial education schemes.

Financial advice

Research on the benefits of financial advice has reached conflicting conclusions. On the one hand, professional advisors have been shown to increase retail investors' performance by improving diversification and reducing disposition effects. On the other hand, returns of advised portfolios did not exceed passive benchmark returns (cf. Stolper and Walter, 2017). Using a large-scale audit study, Mullainathan et al. (2012) show that advisors fail to improve investment behavior because doing so is often not in their best interest. Even if the advice given were unbiased, financial behavior could only be improved if the advice is sufficiently implemented. In their field study of a large-scale financial advice scheme conducted by a German brokerage, Bhattacharya et al. (2012) show that even though investors substantially benefitted from the advice if they followed it, only 5 percent of the sample considered the free and unbiased advice. Among those who considered the advice, more than two-thirds did not follow it at all. Those who did follow the advice did so only to a small extent. As a result, the advice scheme could not enhance investors' portfolios on average. This is particularly relevant given that studies have documented that advice taking and following are positively related to wealth and financial literacy (Bucher-Koenen and Koenen, 2015; Calcagno and Monticone, 2015). This finding implies that financial advice fails to reach those investors who would most benefit from it (Bhattacharya et al., 2012).

Recently, an alternative to traditional financial advice has emerged in the form of robo-advisors. With lower minimum investment amounts and advisory fees, these services can provide less wealthy clients, who tend to be less financially literate (Lusardi and Mitchell, 2014), with access to financial advice (Philippon, 2016). Robo-advisors elicit their clients' risk preferences through standardized online risk questionnaires, based on which portfolio weights are assigned to a set of investment products. The resulting portfolio is continuously monitored and adjusted. Providers mainly rely on portfolio construction and rebalancing algorithms as part of their decision-making process (Jung et al., 2018; Rühr et al., 2019). Bhattacharya et al. (2012) show that a portfolio based on machine-generated advice in the form of a simple and unbiased mean-variance optimizer - as applied by most robo-advisors - can outperform self-managed portfolios by far. While robo-advisors might be able to overcome some of the limitations of traditional advisors, the algorithmic nature of decision-making might inhibit the use of such services. Information systems research has established that decision-making algorithms are often met by skepticism. In particular, decision-makers have been shown to punish errors made by an algorithmic decision support tool more gravely than errors made by a human (Dietvorst et al.,

2015). Thus, even though robo-advisors may improve financial outcomes, the challenge is to overcome investor skepticism regarding the decision-making algorithms.

Research question

Financial education and professional financial advice have been proposed as substitute measures to improve retail investors' financial behavior. However, several studies document that the extent to which advice is sought and followed is related to an investor's level of financial literacy. Thus, we study the effect of a financial education intervention on the use of financial advice. Specifically, we use a randomized control trial, in which participants can choose to delegate all or parts of their experimental budget to a robo-advisory tool. This design allows us to a) identify a potential benefit of financial education not considered so far, namely increased delegation to an unbiased advisory tool, b) overcome endogeneity issues of financial advice studies using measured financial literacy, and c) observe potential spillover effects to individual investment behavior.

We use robo-advisors as a case for three reasons. First, compared to traditional financial advisors, robo-advisors can target less wealthy and less financially literate investors, who would benefit the most from an improvement in financial behavior. Second, they can provide customized information at the point of financial decision-making, which is considered important for the effectiveness of financial education (cf. Willis, 2011; Fernandes et al., 2014). Third, there are calls for mandatory disclosure requirements regarding the applied algorithms, which are conceptually similar to a financial education intervention.³ Specifically, the aim is to improve clients' understanding of the robo-advisors' decision-making process (VZBV, 2018; SEC, 2017). Since decision-making in robo-advisors is mostly derived from modern portfolio theory, increasing the transparency of the inner workings of the algorithms necessarily includes communicating the underlying financial concepts.

While financial advisors rely on information asymmetry as part of their business model, several arguments put forth in related literature point toward a positive impact of financial education on the use of their services. Yaniv and Kleinberger (2000) offer a framework for advice taking in which decision-makers weight conflicting opinions based on the subjective assessment of an

³ A screening of the German robo-advisory market revealed that providers currently do not proactively disclose the underlying reasonings process to the investor. Online-Appendix A.1 displays a summary of the results of the market study.

opinion's merit. Within this framework, decision-makers have privileged access to the reasoning underlying their own opinion, which leads them to place more weight on their own opinion than on an advisor's opinion, a behavior the authors call *egocentric discounting*. Thus, increasing the investor's understanding of the advisor's reasoning process could increase confidence and trust in the advisor's opinion, which has been linked to participation in risky assets and pension plans (Guiso et al., 2008; Agnew et al., 2012). This might be particularly relevant for robo-advisors, who rely on algorithms to make decisions that are crucial to their service: Dietvorst et al. (2015) show that errors made by an algorithmic decision support system are punished more gravely than errors made by a human advisor. Making investors familiar with the investment decision process by opening the "black box" of algorithmic financial decision-making could thus be of greater importance for robo-advisors than for human advisors. Studies from the information systems domain support the notion of a positive effect of explanation facilities on trust and technology adoption (Gregor and Benbasat, 1999; Wang and Benbasat, 2007).⁴ In particular, the provision of transparency seems to increase the appreciation of a robo-advisor (Rühr, 2020). Beyond that, robo-advisors predominantly invest in passively managed index funds, stressing the importance of risk diversification and applying a cost-cutting, low-turnover portfolio strategy. To convince clients of the superiority of this strategy compared to less complex financial products such as a bank deposit or life insurance contract, as well as more complex strategies involving technical and fundamental analyses, it could prove beneficial for them to educate clients on the principles of modern portfolio theory.

3. Experimental design

We conducted an incentivized student experiment in a controlled laboratory environment. In line with the results from a power test we conducted using the variance estimates from a pre-test, we targeted 200 participants. The experiment is structured in the following way. Participants first perform a real effort task that pays a performance-based budget. They are then asked to participate in an incentivized four-period financial decision-making game. In each period, participants allocate their previously earned budget to three distinct assets that evolve probabilistically in value. The value of their portfolio after the last period determines their payout. Participants can delegate decision-making to a robo-advisor for all or parts of their

⁴ A survey conducted by the Bertelsmann foundation corroborates this, as respondents' self-proclaimed knowledge of and attitude towards algorithms is positively correlated (Grzymek and Puntschuh, 2019).

budget at a proportional fee. The service elicits the participants' risk preferences through a risk questionnaire, assigns a risk profile, implements a corresponding portfolio, and reestablishes the portfolio weights after each trading period.⁵ Participants in the treatment group are provided with detailed information on key financial concepts on which the robo-advisory algorithm's decision-making is based, which allows us to investigate the effect of financial education on the propensity to delegate to the robo-advisor.

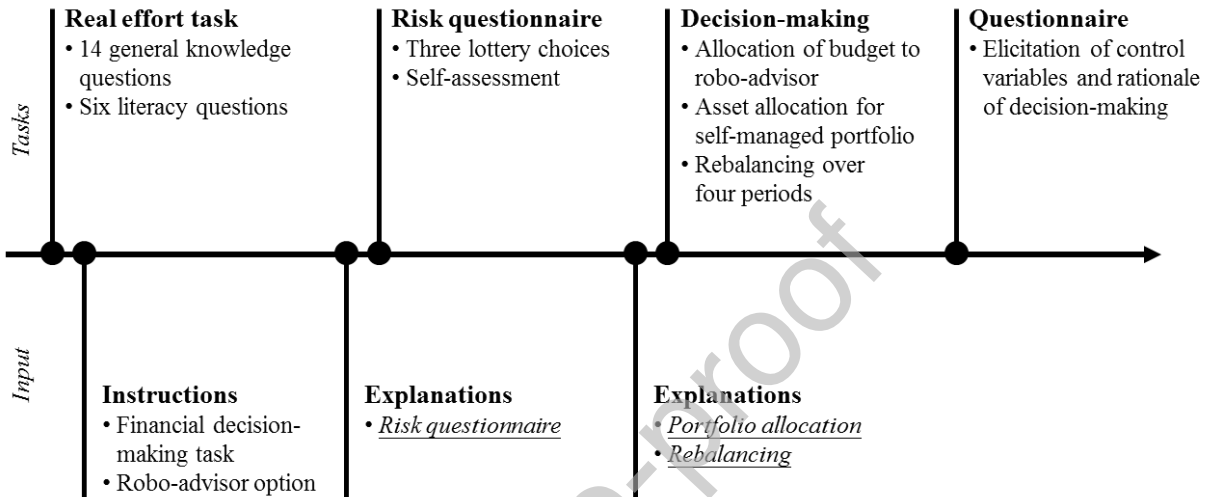


Figure 1: Experimental procedure

Figure 1 illustrates the chronological order of the experimental procedure. The various tasks the participants are required to conduct are depicted in the upper half of the figure. Information that is provided to the participants is depicted in the lower half. The highlighted explanations containing detailed information on the reasoning process of the robo-advisor are only provided to the treatment group. To avoid priming effects and collect all participants' risk preferences, we require both the control and the treatment group to complete the risk questionnaire. The initial decision to allocate parts of the budget to the robo-advisor cannot be reversed, which is intended to prevent participants from withdrawing their robo-advisor share after the first period and replicating the revealed asset allocation in the remaining periods. Participants observe the realized payoffs of the single assets and can adjust their self-managed portfolio allocation each period. The following sections will outline in more detail the procedure within the different parts of our experiment and the rationale behind the design choices.

⁵ See Online-Appendix A.5 for the final payoff distributions of the portfolios implied by the robo-advisor's risk profiles.

General knowledge and literacy questionnaire

At the beginning of the experiment, participants are required to complete a real effort task consisting of a set of 20 general knowledge questions from different domains. Participants receive a payoff of 8€ only if they answer at least ten questions correctly.⁶ However, participants correctly answering more than ten questions do not receive any additional payoffs. This design prevents biases from both windfall endowments and wealth effects in the later stages of the experiment (Keasey and Moon, 1996; Ackert et al., 2006). Among the 20 questions making up the real effort task, six questions allow us to elicit the participants' knowledge of financial (three questions) and technological concepts (three questions).⁷ Including the literacy questions in the effort task allows us to elicit the ex-ante literacy levels of the participants while avoiding a priming effect that would bias behavior in the later stages of the experiment. The order of these questions is randomized to avoid order effects.⁸

Financial decision-making task and robo-advisory option

Participants that have completed the real effort task can increase their payoffs in repeated asset allocation tasks inspired by a related study's design (cf. Kuhnen, 2015), in which participants allocate their budget to three distinct assets. Asset *A* is a risk-free asset paying 5 percent with certainty. Asset *B* is a stock of moderate risk, paying +21 percent or -7 percent with equal probability. Asset *C* is a stock of high risk, paying +35 percent or -21 percent with equal probability. The realization of the risky assets' payoffs is mutually independent. Thus, asset *C* yields the same expected return as asset *B* at higher volatility, which might convey the impression that asset *C* is inferior to asset *B*. However, investors can reduce the volatility of their risky portfolios by combining assets *B* and *C* while keeping constant the expected rate of return. As a result, the minimum-variance portfolio constructed from the two risky assets dominates any other combination of the two risky assets for any risk-averse participant.⁹ The robo-advisor identifies the minimum-variance portfolio of the two risky assets and combines it with the risk-free asset *A* to accord with the individual risk preferences of a participant. Thus,

⁶ Participants who answer less than ten questions correctly are awarded 4€ and allowed to carry on participating in the experiment. In our sample, 196 of 205 participants received the high payment.

⁷ We designed questions to capture a respondent's level of technological literacy. Since these questions have not yet been verified, implications should be treated with caution.

⁸ A list containing all questions can be found in Online-Appendix A.7.

⁹ The minimum-variance portfolio contains 80 percent of asset *B*, and 20 percent of asset *C*. Calculations are provided in Online-Appendix A.2.

using the robo-advisor prevents investors from investing in a dominated risky portfolio. However, since the degree of risk aversion determines the utility-maximizing budget shares of the risk-free asset and the risky minimum-variance portfolio, there is no globally optimal portfolio. While related research on financial decision-making has relied on tasks that have some form of optimal strategy (Kuhnen, 2015; Jung & Weinhardt, 2018), we are interested in the tendency of participants to delegate financial decision-making to a robo-advisor that can a) calculate the portfolio weights of the optimal risky portfolio, which prevents participants from holding a dominated risky portfolio, b) elicit the participants' risk preferences through a questionnaire and assign corresponding shares to the risk-free asset and the optimal risky portfolio, and c) reestablish the original allocation in response to changes in the assets' values. The robo-advisor unambiguously provides value in the first task since the exact shares are hard to assess, even for financially literate individuals. The same does not hold for the second task since participants might be better suited to assess their risk preferences than the robo-advisor (cf. Tertilt & Scholz, 2018).

To make sure that using the robo-advisor is not a dominated option, we calibrated the maximum fee participants would be willing to pay by computing the additional utility that the use of the service yields in comparison to three counterfactual scenarios (worst possible investment strategy, risk-free portfolio strategy, and 1/N diversification strategy) and for each of the four risk profile groups. The results suggest that the maximum fee per period ranges between 0.4 percent and 12.1 percent of managed assets in the experiment¹⁰, which is a realistic reference point considering that actual fees range from 0.4 percent to 1.2 percent in the German market (Stiftung Warentest, 2018).¹¹ Our pre-test confirmed that a per-period fee of 0.5 percent achieved sufficient variation in usage in the baseline case.

The financial decision-making task lasts four periods. After each period, participants observe the realized payoffs and have the opportunity to change the composition of their self-managed asset allocation. Before trading starts, participants have the option to allocate parts of their budget to the robo-advisory algorithm at a per-period fee of 0.5 percent of the assets assigned to the robo-advisor. The allocation to the robo-advisor cannot be reversed in the later periods. The design of the robo-advisor is explained to the participants in the form of an animated video, which enhances participants' comprehension. Besides, it avoids announcer effects and simplifies the replication of the experiment by standardizing the intervention. While the

¹⁰ Calculations are based on a $\mu - \sigma$ utility function and displayed in Online-Appendix A.6.

¹¹ The fees are based on a portfolio value of 51,000€.

instructions regarding the decision-making task, as well as essential information regarding the three core process steps of the robo-advisor, are shown to both treatment and control group, only participants in the treatment group receive detailed information on the financial intuition and rationale behind the various process steps (risk profiling, portfolio recommendation, rebalancing).

The robo-advisor used in this experiment mirrors the most frequently observed features of current providers (Rühr et al., 2019). These features include eliciting risk preferences through a brief risk questionnaire, implementing a corresponding portfolio, and ongoing rebalancing. The risk questionnaire requires each participant to choose among a risk-free payoff and a risky lottery three times. The risky lottery choice becomes more appealing with each choice, while the risk-free option remains the same throughout the questionnaire.¹² Thus, the earlier a participant switches from the risk-free option to the risky lottery, the higher that participant's risk tolerance. The robo-advisor's decision-making process takes as inputs the answers to three lottery choices each specifically designed to tease out whether a participant's risk aversion coefficient is above or below a certain threshold. A utility function is fitted to these coefficients, which allows assigning each participant (provided that the reported preferences are consistent) one of four risk profiles spanning distinct ranges of risk aversion. Based on the assigned risk profile, a suitable portfolio that reflects the risk preferences of the participants is implemented and rebalanced by the robo-advisor.¹³ In addition to this quantitative elicitation measure, we gather a qualitative estimate, which is commonly used in the field.¹⁴

Treatment

In line with the regulation envisioned by the German consumer protection agency and the SEC, we manipulate the instructions given to experimental participants by providing detailed information regarding the rationale behind the decision-making algorithm to the treatment group. Since decision-making in robo-advisors is based on modern portfolio theory, this necessarily includes communicating the underlying financial concepts. The additional information is provided in the form of *how* explanations that were initially proposed by Wang

¹² A detailed account of the lottery choices is contained in Online-Appendix A.3.

¹³ Detailed calculations regarding the derivation of the recommended portfolios, as well as the math behind the lottery choices and portfolio recommendations are provided in Online-Appendices A.1, A.3, and A.4.

¹⁴ "How do you see yourself: Are you generally a person who is fully prepared to take risks or do you try to avoid taking risks?", cf. Dohmen et al., 2011

and Benbasat (2007). These explanations “reveal the line of reasoning [...] based on consumer needs and product attribute preferences, and they outline the logical processes involved in reaching final recommendations” (p. 224). The explanations are provided to the participants in the form of an animated video to exploit the digital nature of the service provision and increase the degree of understanding (Lusardi et al., 2015). As opposed to the treatment group, participants of the control group only receive essential information regarding the three core process steps of the robo-advisor, which is in line with the current market situation (cf. Online-Appendix A.1). We carefully designed the instructions to provide all the information required to understand the robo-advisor’s service to both the control and the treatment group. This design ensures that differences in delegation do not result from differences in the perceived extent of the service.

The explanations follow the three process steps of the robo-advisory tool (risk questionnaire, portfolio allocation, and rebalancing). The risk questionnaire explanation illustrates that higher returns are typically associated with higher risk and that an optimal portfolio is always subject to an investor’s risk preferences. Further, the mechanics of the robo-advisor’s risk questionnaire are explained using an example, in which the relation between the participant’s answers and the assigned risk profile is highlighted. The portfolio allocation explanation provides participants with an intuitive introduction to modern portfolio theory, explaining the concept of diversification and introducing the optimal market portfolio. Participants are demonstrated how their risk preferences affect the combination of the market portfolio with the risk-free asset. The rebalancing explanation demonstrates that the initial portfolio allocation changes after the asset values are realized in each period. The robo-advisor realigns the portfolio with a participant’s risk preferences by establishing the initial allocation.

The volume and complexity of information provided to the treatment group have been discussed as a potential threat to the identification of the treatment effect (Bergstrom and Stoll, 1990). To avoid differences in cognitive exhaustion, we show the control group an unrelated video of similar length and style in addition to the (shorter) instruction video. The multimedia presentation format of the instructions should decrease the degree of attention needed to process a given amount of information. After each video, participants answer a set of comprehension questions. In the treatment group, the aim is to a) test the effectiveness of the education treatment, b) stimulate attention, and c) obtain a control variable signaling the degree of comprehension in the treatment dimensions, i.e., the process steps of the robo-advisor. The control group receives the same amount of questions of a similar nature, which ensures

comparability by equally stimulating attention and thereby inducing the same cognitive exhaustion.

Control variables

We collect several control variables in addition to standard demographic variables (such as gender and age) to capture individual characteristics that might affect the delegation decision. Firstly, overconfidence is associated with costly investment behavior (Barber and Odean, 2000), as well as a lower likelihood of seeking financial advice (Guiso and Jappelli, 2006; Kramer, 2016) and delegating financial decisions (Calcagno and Monticone, 2015). Thus, we construct an overconfidence measure by asking participants how many questions they believe they answered correctly in the effort task and subtracting from that the number of questions they in fact answered correctly. Secondly, more risk-averse individuals tend to be more likely to seek advice (Bluethgen et al., 2008). We elicit a participant's risk preferences using three distinct measures. The first measure involves a lottery choice task. The second measure is based on self-assessment on an established scale (cf. Dohmen et al., 2011). The third measure is based on participants' behavior in the financial decision-making task. Specifically, we use the excess portfolio share allocated to the risk-free asset A in the self-managed portfolio portion to obtain a revealed measure of risk aversion.¹⁵ Since financially illiterate individuals stand to gain most from the delegation of portfolio decisions (Bianchi, 2018; Agarwal et al., 2017), we construct a financial literacy measure from the three questions testing financial knowledge in the effort task (cf. van Rooij et al., 2011), as well as a measure based on self-assessment. Aside from financial literacy, investors of robo-advisors might need a sufficient technological understanding of the process of algorithmic decision-making. Research has shown that the use of intelligent systems positively affects the use of a different yet similar system (Venkatesh and Davis, 1996; McFarland and Hamilton, 2006). Thus, we construct a technological literacy measure from the three questions testing technological knowledge in the effort task, as well as a measure based on self-assessment. We further inquire about the attitude toward new technologies (*technology readiness*, c.f. Parasuraman, 2000).

¹⁵ We define the excess portfolio share of the risk-free asset as the observed portfolio share of asset A in the self-managed budget share minus the optimal portfolio share of asset A as inferred from the assumed utility function.

4. Data

Prior to conducting the experiment, we extensively pre-tested the design with 16 individuals of different backgrounds. After the design was finalized, we pre-registered the experiment on the *American Economic Association's* registry for randomized controlled trials.¹⁶ The experiment was conducted in ten sessions (May 20th to May 22nd) at the MELESSA laboratory of the LMU Munich (Germany). We recruited 230 participants using the ORSEE software (Greiner, 2015), of which 205 showed up and completed the experiment. Randomization was conducted on two levels: participants were randomly assigned a computer in the lab, and each computer was randomly assigned to either the treatment or the control group. As a result, the treatment group consisted of 105 participants, and the control group consisted of 100 participants.

Table 1: Summary statistics

	Mean	St. dev.	Min.	p25	p50	p75	Max.
Duration (seconds)	1614.56	343.68	941	1363	1560	1784	2875
Age (years)	26.04	9.34	16	21	23	27	70
Female (D)	0.60	-	-	-	-	-	-
Student (D)	0.85	-	-	-	-	-	-
Overconfidence	-2.98	2.88	-11	-5	-3	-1	6
Risk aversion (qualitative)	4.98	2.41	0	3	5	7	10
Risk aversion (quantitative)	0.76	1.09	0	0	0	1	3
Financial literacy (elicited)	1.76	0.83	0	1	2	2	3
Financial literacy (self-reported)	3.10	2.43	0	1	3	5	10
Technological literacy (elicited)	1.90	1.00	0	1	2	3	3
Technological literacy (self-reported)	4.20	2.74	0	2	4	6	10
Technology readiness	3.93	2.26	0	2	4	6	10
Effort score	15.27	2.49	7	14	15	17	20
Budget share delegated	0.32	0.29	0.00	0.02	0.27	0.50	1.00
Final payoff (€)	15.15	2.24	8.77	14.15	15.02	16.23	22.15
Return on budget	0.30	0.25	-0.33	0.16	0.26	0.42	1.14

Note: This table reports summary statistics for the main characteristics of the participants. The number of observations was 205 for all variables. Note that the final payoff includes a fixed show-up fee of 5€.

¹⁶ Litterscheidt and Streich (2019)

Table 1 reports summary statistics for the main characteristics of the participants. The median participant took 26 minutes to complete the experiment. On average, participants were 26 years old, with a quarter of the observations exceeding 27 years. Sixty percent of participants were female, and 85 percent were enrolled as students at the time of the experiment. More than three-quarters of the participants underestimated the number of questions they answered correctly in the effort task and were classified as underconfident. Both the qualitative and the quantitative risk aversion measures vary substantially, with the former somewhat evenly distributed around the median and the latter positively skewed. The elicited financial and technological literacy variables are defined as the number of correctly answered questions from the two domains in the effort task (for exact specifications see Online-Appendix A.7). The self-reported literacy and readiness values are measured on an 11-point Likert scale. On average, participants answered 15.27 of the 20 general knowledge questions correctly, with nine participants falling below the threshold of ten correct answers and thus receiving an initial budget of €4. The primary variable of interest is the share of the budget delegated to the robo-advisor. On average, participants delegated 32 percent of their initial budget to the robo-advisory tool. The average payoff to participants was €15.15, which comprises a fixed show-up fee of €5 and an average return on the initial budget of 30 percent.

We conducted non-parametric Mann-Whitney tests to identify any significant deviations in the observed variables between the treatment and control groups. Whenever significant differences were found (which was the case for age and duration), we included the variables as control variables in the subsequent analyses. The experiment was conducted on three consecutive days. To ensure that outliers do not drive our results, we compare the distributions of the variable of interest and show that there is no significant difference between the three days.¹⁷

As the budget share delegated to the robo-advisor is our primary dependent variable, Online-Appendix A.8 contains descriptive statistics and illustrations of the distribution of the delegated budget share. We observe that 55 participants allocated less than ten percent to the robo-advisor, including 50 participants that managed their entire budget themselves. As suggested by a Saphiro-Wilk test, normality of the observed sample is rejected. Thus, we conduct non-parametric tests throughout the following analysis to account for the non-normality and right-skewness of the distribution.

¹⁷ A summary of the tests is displayed in Online-Appendix A.8.

5. Results

The main question we try to answer in this study is whether the willingness to use a robo-advisor is affected by a financial education intervention. We show that participants in the treatment group delegate a significantly higher share of their budget to the robo-advisor than participants in the control group. Figure 2 displays the univariate treatment effect, which is highly significant and amounts to 15.26 percentage points when considering the entire sample and 20.12 percentage points when considering only the subset of participants that have correctly answered all comprehension questions. Considering an average share of 32.3 percent in the entire sample and 24.5 percent in the control group, this represents an economically significant effect.

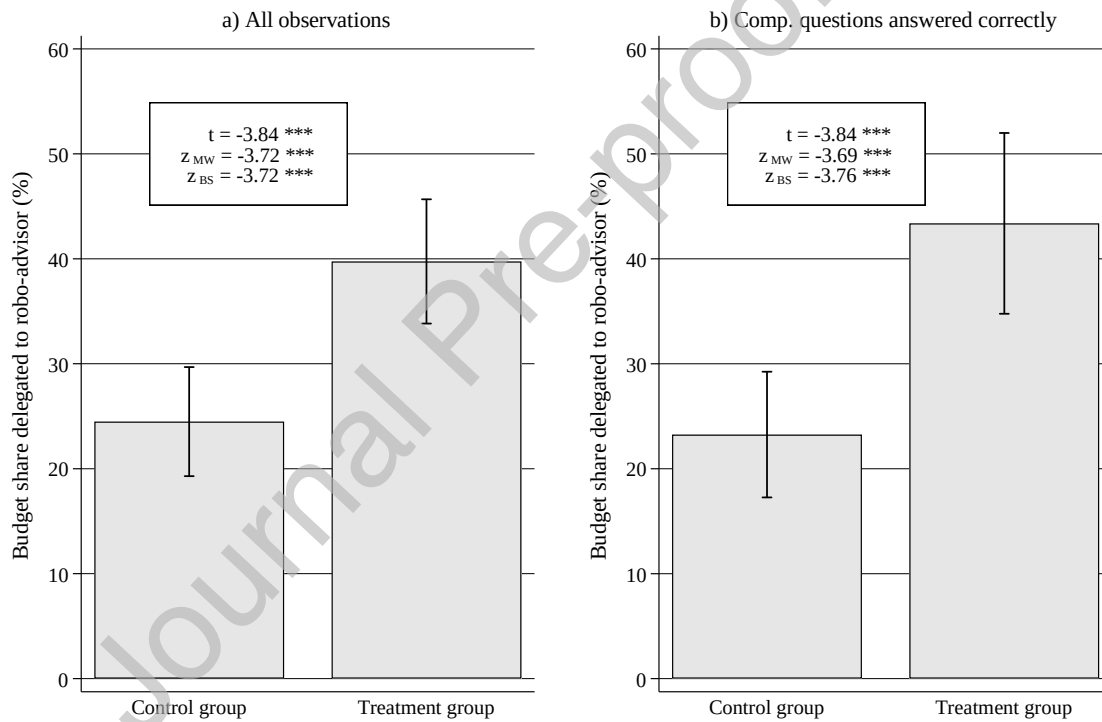


Figure 2: Main treatment effect (univariate)

Note: This figure displays the average budget share delegated to the robo-advisory tool and the 95 percent confidence interval by groups for a) the entire sample ($N = 205$) and b) participants that have correctly answered all comprehensions questions ($N = 121$). Test statistics are reported for two-sample t-tests (t), non-parametric Mann-Whitney tests (z_{MW}), and parametric estimates on 9,999 bootstrap samples (z_{BS}).

We report the results of a more detailed empirical analysis in Table 2. We perform OLS regressions with and without control variables (columns 1 and 2), as well as a fixed effects regression to account for session-specific characteristics (column 3):

$$Delegation_{i,s} = \beta_0 + \beta_1 Treatment_i + \beta_2 \gamma_i + \vartheta_s + \varepsilon_i \quad (1)$$

where $Delegation_{i,s}$ is the budget share participant i delegated to the robo-advisor, $Treatment_i$ is a dummy variable indicating treatment, γ_i includes participant-level control variables, ϑ_s includes session fixed effects, and ε_i is the error term. The treatment coefficient remains stable across specifications and implies that educating participants on the inner workings of the robo-advisory algorithm increases the budget share delegated to the service by more than 15 percentage points.

Table 2: Multivariate analysis of treatment effect

	(1) <i>Delegation</i>	(2) <i>Delegation</i>	(3) <i>Delegation</i>
<i>Treatment</i>	0.153*** (0.040)	0.156*** (0.041)	0.162*** (0.042)
<i>Constant</i>	0.245*** (0.026)	0.287** (0.115)	0.261* (0.140)
Obs.	205	205	205
Controls	No	Yes	Yes
Session fixed effects	No	No	Yes
R2	6.73%	7.05%	8.33%
Adj. R2	6.28%	4.72%	1.57%
F-value	14.77	3.021	1.221
p-value	0.00	0.01	0.26
d.f.	203	199	190

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Note: This table displays the coefficients of regressions on the budget share participants delegated to the robo-advisor. In columns 2 and 3, we control for age, duration, financial literacy, and technological literacy. In column 3, we include session fixed effects to account for session-specific characteristics. The treatment coefficient is robust to the inclusion of clustered standard errors at the session-level.

In the next step, we investigate how the treatment affected participants' behavior in the financial decision-making task. Since the treatment group is provided with detailed information

regarding the inner workings of the robo-advisor in the three process steps (risk profiling, portfolio allocation, and rebalancing), we test whether the treatment directly affects participants' behavior in these three steps. As most participants decided to manage at least a part of their budget themselves (193 of 205), we can exploit their behavior in the financial decision-making task. More specifically, we can examine whether participants in the treatment group behave differently from participants in the control group in terms of a) their risk preferences, b) portfolio allocation, and c) rebalancing.

Risk preferences

We gather information regarding the participants' risk preferences in the form of a risk questionnaire designed to resemble risk questionnaires applied in the field. In addition to this quantitative measure of risk aversion, we asked participants to provide an assessment of their risk preferences using an established scale (cf. Dohmen et al., 2011). Exploiting our experimental design, we compute a third measure that – as opposed to the two *stated* measures – captures a subject's *revealed* risk preferences. We use the excess share of the self-managed budget portion allocated to the risk-free asset A in the first period of the financial decision-making game as a measure of risk aversion. According to modern portfolio theory, an investor's risk preferences should govern the share of wealth invested in the risk-free asset. Thus, using the share invested in the risk-free asset in the first period ($w_{A,i}^{t=1}$) allows us to infer a subject's level of risk aversion.¹⁸ We subtract from that measure the optimal share of the risk-free asset A according to the individual risk profile elicited in the risk questionnaire ($w_{A,i}^*$, see Online-Appendix A.4):

$$RiskAversion_{excess,i} = w_{A,i}^{t=1} - w_{A,i}^* \quad (2)$$

$RiskAversion_{excess,i}$ is a measure of a participant's *excess* risk aversion over the elicited preferences. If this measure is larger than (smaller than) zero, the participant's allocation reveals a higher (lower) level of risk aversion than suggested by her risk profile.

Next, we investigate whether the three risk aversion measures systematically differ between participants in the treatment and control group. Table 3 reports summary statistics for the three measures by groups, as well as test statistics for non-parametric Mann-Whitney tests. The tests

¹⁸ We use information from the first period to avoid distortions from wealth effects (cf. Paravisini et al., 2017) and from feedback regarding return realizations (cf. Gneezy and Potters, 1997).

suggest that participants in the treatment group do not differ from participants in the control group in terms of the quantitative risk aversion measure. However, their self-reported risk aversion is significantly lower than that of the control group. At the same time, treated participants' behavior in the financial decision-making game suggests that their excess risk aversion is significantly higher than that of the control group.

Table 3: Risk aversion measures by groups

	Control		Treatment		z score
	Mean	St. dev.	Mean	St. dev.	
Risk aversion (quantitative)	0.76	1.28	0.43	1.00	1.42
Risk aversion (qualitative)	5.24	2.55	4.25	2.33	2.41 **
Risk aversion (excess)	20.42	29.01	28.27	33.18	2.41 **

Note: This table displays means and standard deviations of three risk aversion measures by groups. The quantitative measure is derived from the answers to the lottery choices in the risk questionnaire (0 – *least risk-averse* to 3 – *most risk-averse*). The qualitative measure is based on a self-assessment of participants on an 11-point Likert scale (0 – *least risk-averse* to 10 – *most risk-averse*). The excess risk aversion is based on revealed preferences of participants in the allocation stage. The last column contains the z scores of non-parametric Mann-Whitney tests. Significance levels are indicated by stars (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). The sample is restricted to subjects that have answered the lottery choices consistently and – in the case of the treatment group - have correctly answered the comprehension question targeting the risk profiling explanations in the treatment video (N = 154).

Since two of the univariate differences are statistically significant, we further investigate the relationship between the various risk aversion measures. First, we analyze whether the difference in the qualitative risk aversion measure persists when we account for a participant's quantitative risk aversion measure. Thus, we report the coefficients of a multivariate linear regression on the qualitative risk aversion measure in Online-Appendix A.8, which imply that the treatment decreases a participant's self-reported risk aversion even when controlling for the quantitative risk aversion measure. These results are consistent with an anchoring effect of the risk questionnaire on the self-assessment in the treatment group, which is induced by the risk profile explanation.

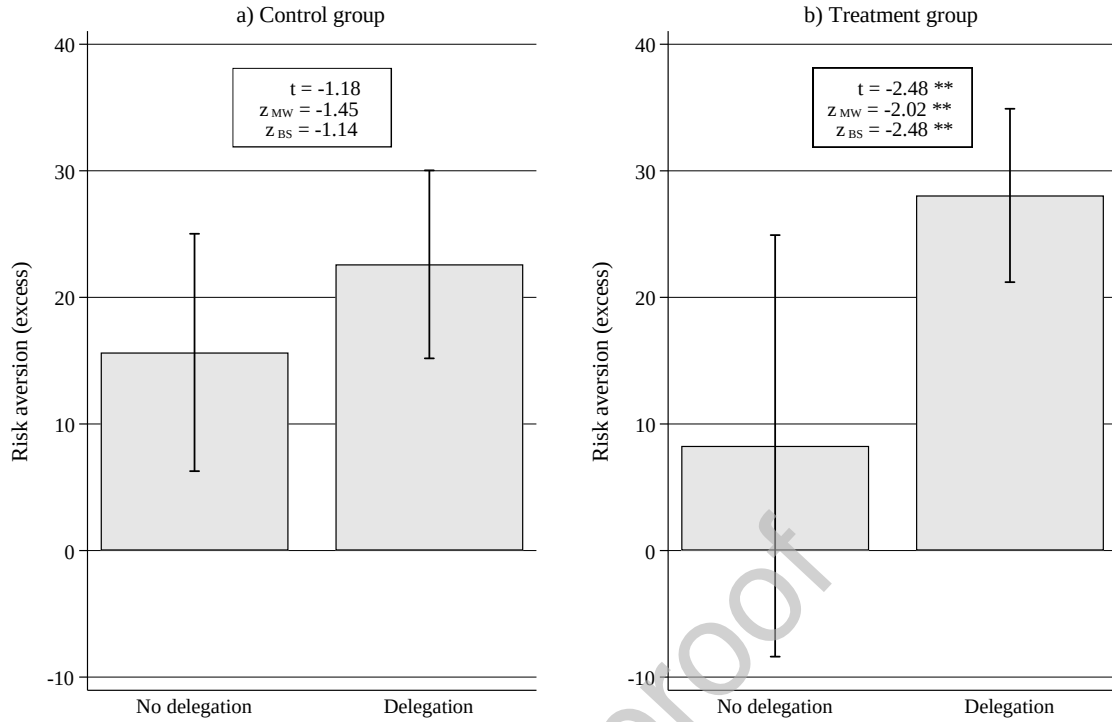


Figure 3: Comparison of excess risk aversion

Note: This figure displays the average excess risk aversion (i.e., the difference between budget share invested in the risk-free asset A in period 1 and budget share suggested for respective risk profile) and the 95 percent confidence interval by delegation to the robo-advisor for a) the control group (N = 98) and b) the treatment group (N = 95). In the control (treatment) group, 65 (78) participants delegated at least a part of their budget. The sample was restricted to participants that have not delegated their entire budget to the robo-advisor and – in the case of the treatment group – correctly answered the comprehension question targeting the risk profile explanations in the treatment video. Test statistics are reported for two-sample t-tests (t), non-parametric Mann-Whitney tests (z_{MW}), and parametric estimates on 9,999 bootstrap samples (z_{BS}).

Second, we take a closer look at the excess risk aversion measure. Recall that participants can delegate part of their budget to the robo-advisor and manage the remaining part themselves. The values for the excess risk aversion imply that the average participant in both the control and treatment group acted more risk-averse than suggested by the robo-advisor's risk profile. This difference is significantly higher in the treatment group than in the control group (see Table 3). Within the treatment group, participants that use the robo-advisor display higher levels of excess risk aversion than participants who do not use the robo-advisor (see Figure 3).

Counterintuitively, the risk profile explanation seems to have opposing effects on the participants' self-reported and revealed levels of risk aversion. On the one hand, the self-

reported level of risk aversion decreases. On the other hand, the participants' portfolio choices in the self-managed budget share suggest an increase in risk aversion. Both effects can be rationalized by increased awareness of how the risk questionnaire affects the risk profile of the portfolio suggested by the robo-advisor.

Portfolio allocation

Portfolio diversification is a central feature of modern portfolio theory. The idea of combining non-perfectly correlated risky assets to decrease asset-specific risk is essential to maximize risk-adjusted portfolio returns. In our experiment, we designed the two risky assets such that both have the same expected return, while one risky asset has a lower standard deviation than the other risky asset. This construction might convey the impression that the latter is inferior to the former. However, since the realizations of the two assets' returns are independent, any combination of the two risky assets is superior to both assets in terms of its risk-adjusted return. Consequently, we would expect participants that have understood the principles of diversification to invest in both risky assets instead of either one. Thus, we construct our measure of diversification as the average share of periods in which the participants diversified their risky portfolio, i.e., periods in which they invested in both risky assets. Figure 4 illustrates the diversification variable for the control and treatment group.

The results of parametric and non-parametric tests indicate that, on average, participants of the treatment group diversify more often than participants of the control group. This behavior indicates that the portfolio allocation part of the treatment is effective since participants make decisions that are in line with modern portfolio theory. Thus, educating participants about portfolio diversification indeed improved their financial decision-making ability. However, this raises the question as to whether educated investors should delegate investment to a robo-advisory tool at all.

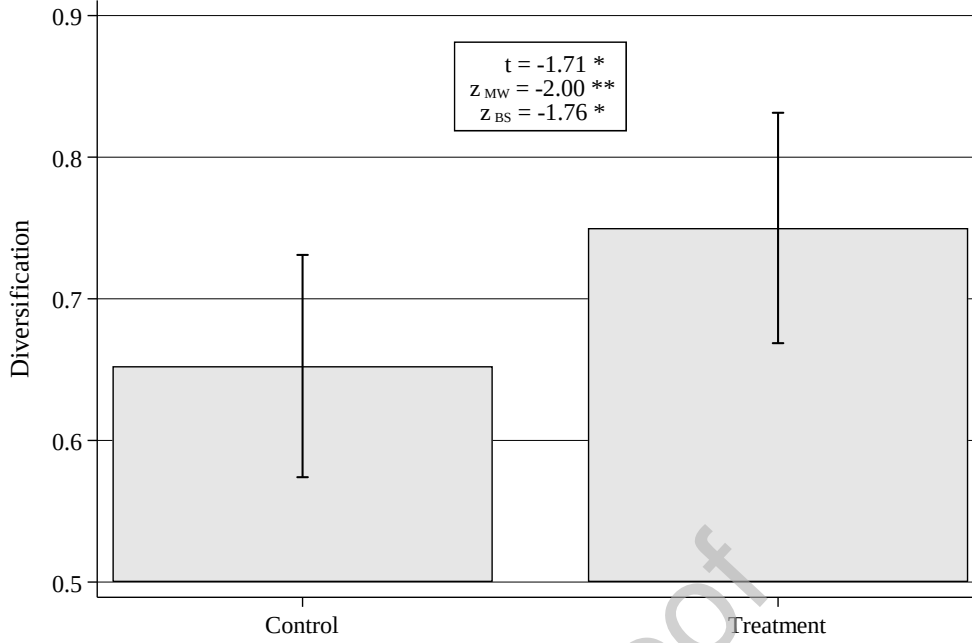


Figure 4: Treatment effect on diversification (univariate)

Note: This figure displays the average fraction of periods in which the participants included both risky assets in their portfolios by groups. The sample was restricted to participants that have a) invested in at least one risky asset and b) – in the case of the treatment group - correctly answered the comprehension question targeting the portfolio allocation explanations in the treatment video (N = 183). We report test statistics for two-sample t-tests (t), non-parametric Mann-Whitney tests (z_{MW}), and parametric estimates on 9,999 bootstrap samples (z_{BS}).

To investigate whether enhanced diversification displaced adoption, we run a multivariate regression of the tendency to delegate to the robo-advisor on the treatment and the diversification measure. Table 4 displays the results of OLS regressions with and without control variables (columns 1 and 2), a fixed effects regression to account for session-specific characteristics (column 3), as well as a regression using a restricted sample (column 4):

$$Delegation_{i,s} = \beta_0 + \beta_1 T_i + \beta_2 Diversification_i + \beta_3 \gamma_i + \vartheta_s + \varepsilon_i \quad (3)$$

where $Diversification_i$ is a measure ranging between zero and one. The diversification coefficients are statistically significant and range from 10.31 percentage points to 13.16 percentage points, which represents an economically significant relationship. The coefficient remains stable across specifications and implies that participants that more frequently diversify their portfolio delegate a higher share of their budget to the robo-advisor.

Table 4: Multivariate analysis of diversification on delegation

	(1) <i>Delegation</i>	(2) <i>Delegation</i>	(3) <i>Delegation</i>	(4) <i>Delegation</i>
<i>Treatment</i>	0.138*** (0.039)	0.141*** (0.040)	0.145*** (0.041)	0.167*** (0.045)
<i>Diversification</i>	0.132*** (0.047)	0.129*** (0.047)	0.128*** (0.048)	0.103** (0.051)
<i>Constant</i>	0.159*** (0.038)	0.195* (0.111)	0.179 (0.132)	0.273* (0.142)
Obs.	205	205	205	183
Controls	No	Yes	Yes	Yes
Session fixed effects	No	No	Yes	Yes
R2	9.53%	9.60%	10.70%	11.90%
Adj. R2	8.63%	7.79%	4.57%	5.12%
F-value	10.86	5.394	1.869	1.794
p-value	0.00	0.00	0.04	0.05
d.f.	202	200	191	169

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Note: This table displays the coefficients of regressions on the budget share participants delegated to the robo-advisor. In columns 2 and 3, we control for age, duration, financial literacy, and technological literacy. In column 3, we include session fixed effects accounting for session-specific characteristics. In column 4, we restrict the sample to participants that have a) invested in a risky asset and b) – in the case of the treatment group - correctly answered the comprehension question targeting the portfolio allocation explanations in the treatment video (N = 183). The treatment coefficients are robust to the inclusion of clustered standard errors at the session-level.

Rebalancing

Rebalancing refers to the re-establishment of the initial asset allocation in response to changes in the portfolio weights. Given that the initial allocation adequately reflected a participant's risk preferences and assuming stable preferences, rebalancing ensures that the participants' preferences and the risk profile of their respective portfolios remain aligned. In the experiment, the robo-advisory tool takes care of rebalancing in the delegated budget share, while participants are in charge of the portfolio weights in the self-managed part of their budgets. Since the treatment video explains the underlying reasoning behind the concept of rebalancing, we would expect participants in the treatment group to engage more actively in rebalancing than participants of the control group.

In the context of the financial decision-making game, rebalancing efforts relate to the weights of the two risky assets in the risky portfolio. We do not focus on the fraction of the share of the risk-free asset since this proportion is driven by the risk preferences of the investors and potential wealth effects of prior returns. Thus, we define a novel, inverse measure of a participant's rebalancing ability (*Rebalancing_i*), which is defined as the standard deviation of the share of asset B within the risky portfolio over the four periods:

$$Rebalancing_i = \frac{1}{4} \sum_{t=1}^4 \left(\frac{w_{B,i}^t}{w_{B,i}^t + w_{C,i}^t} - \frac{1}{4} \sum_{t=1}^4 \frac{w_{B,i}^t}{w_{B,i}^t + w_{C,i}^t} \right)^2 \quad (4)$$

where $w_{B,i}^t$ is the share of asset B in participant i 's portfolio in period t , and $w_{C,i}^t$ is the share of asset C in participant i 's portfolio in period t . Participants with a *Rebalancing_i* value equal to zero maintained their initial asset allocation in each period, suggesting high rebalancing ability. Participants with a higher *Rebalancing_i* value on average more frequently or extensively deviated from their average portfolio allocation, suggesting lower rebalancing ability. In Online-Appendix A.8, we compare this measure across both groups to evaluate whether the part of the treatment video referring to rebalancing affected participants' behavior.

The test statistics suggest that the control and treatment groups do not differ in terms of rebalancing ability. Taking a closer look at the distribution of the rebalancing measure, we find that most of the participants barely have any variation in their asset allocation, suggesting high rebalancing ability throughout the sample.¹⁹ The static payoff profile of the assets, as well as the small number of trading periods, might have contributed to stable allocation choices. In addition, the fact that participants do not need to buy and sell assets to reach a particular portfolio allocation, but instead directly state the desired portfolio weights decreases the costs of rebalancing.

6. Discussion

We study financial decision-making in a randomized control trial, contributing to existing literature on the effectiveness of financial education by allowing participants to delegate financial decision-making. Specifically, we conduct our study in the context of robo-advisors that might be in a good position to effectively educate investors due to their broad investor base

¹⁹ See Online-Appendix A.8 for an illustration of the distribution of the rebalancing measure.

and digital service provision, which allows for just-in-time interventions (cf. Fernandes et al., 2014). At the same time, it might be more relevant for robo-advisors than for traditional advisors to communicate their underlying investment logic to oppose algorithm aversion due to the perception of a ‘black box’ decision-making process (Dietvorst et al., 2015).

Our results show that providing detailed information regarding the inner workings of a robo-advisory algorithm increases participants' tendency to delegate financial decision-making. The effects in the laboratory are substantial and stable across several specifications. Aside from the main treatment effect, we identify two insightful effects on the behavior of the treated participants. First, we observe conflicting alterations in different risk aversion measures. Second, treated participants display improved portfolio diversification, which does not displace delegation.

Delegation

We find that the financial education intervention increases delegation to the robo-advisor. Our interpretation of this finding is twofold: First, the intervention may have affected how participants perceive the investment decision. Specifically, educating investors on the considerations involved in the investment decision can have one of two consequences. On the one hand, it may increase investors' understanding of the decision principles, which increases their confidence to make the decision themselves. On the other hand, it may highlight the complexity of the decision and the improvement in the financial outcome that a viable investment strategy can provide. Given that we observe a greater extent of delegation among participants that received the financial education intervention, the latter seems more plausible. Second, the intervention may have affected how participants perceive the robo-advisor's service. Specifically, the intervention may have increased trust in the service as it increased the investor's understanding of the algorithm's reasoning process, which seems to be particularly relevant for algorithmic decision-support systems such as robo-advisors (cf. Dietvorst et al., 2015). Trust may be higher because the treatment was effective at educating participants or because it provided more transparency regarding the service (Rühr, 2020). The design of our treatment, as well as the fact that treated participants were better at diversifying their portfolios, suggests that the effect was not only driven by increased transparency.

Risk aversion

Concerning the risk aversion measures, we identify two opposing effects. On the one hand, we observe lower levels of the self-reported risk aversion measure for treated participants, even when controlling for the risk aversion measure elicited through the risk questionnaire. On average, participants overestimate their level of risk aversion: the median participant's answers to the risk questionnaire were consistent with the riskiest investor profile, while the median participant stated to be moderately risk-averse, i.e., five on a scale from zero to ten. Thus, the treatment seems to have enhanced the participants' understanding of their risk preferences, which is consistent with an anchoring effect stemming from the risk profile explanation.

On the other hand, we observe higher levels of excess risk aversion among treated participants. The difference seems to be driven by treated participants who at least partially delegate decision-making to the robo-advisor. Specifically, we find significantly higher excess risk aversion among delegating participants in the treatment group, but not in the control group. We identify two potential interpretations of the observed patterns. First, participants might answer the risk questionnaire strategically to save fees: Participants incur fees in proportion to the total amount delegated to the robo-advisor. Our treatment aims to increase participants' understanding of how their answers to the risk questionnaire translate into the portfolio implemented by the robo-advisor. Thus, treated participants, who obtain more information about the relationship between their risk profiles and the robo-advisor's portfolio allocation, may only want to use the robo-advisor to manage the risky part of their portfolio (optimal combination of risky assets B and C) and determine the risk-free share of their portfolio (share of risk-free asset A) themselves. This could be achieved by understating risk aversion in the questionnaire and delegating only the risky share of their budget to the robo-advisor. This would be associated with a higher risky share in the delegated portfolio and, correspondingly, a higher risk-free share in the self-managed portfolio, which could explain the observed differences in excess risk aversion. To test this interpretation, we compare the risk profiles derived from the questionnaire between the control and treatment group, as well as between delegating and non-delegating participants within the treatment group. Since the measure does not differ significantly for either comparison, we discard this interpretation.

Second, participants might strategically increase the risk-free share in the self-managed portfolio to counteract the risk of the portfolio implemented by the robo-advisor, which they may perceive to be excessive. We observe a marked difference between participants' self-

reported risk aversion and the risk profiles assigned to them. Participants in the treatment group might be made aware of this deviation, which would cause them to offset the risk in the delegated portfolio by taking less risk in the self-managed portfolio. Two assumptions need to hold for this interpretation to be plausible: First, the treatment improves the awareness of the gap in self-reported and assigned risk aversion. The fact that we observe lower levels of self-reported risk aversion for treated participants (which aligns self-reported and assigned risk aversion) provides some support for this assumption. Second, participants react to the perceived deviation by strategically adjusting the risk-free share in the self-managed portfolio. The fact that excess risk aversion is positively associated with delegation to the robo-advisor in the treatment group provides some support for that assumption.

Diversification

Two findings emerge from analyzing the impact of the portfolio allocation part of the treatment. On the one hand, we find that treated participants more frequently diversify their risky portfolios, which suggests that the treatment enhances the understanding of the principles of portfolio diversification. On the other hand, this improvement does not contest the use of the robo-advisor. Instead, participants with higher diversification abilities were more inclined to delegate to the robo-advisor. This finding is at odds with the results of previous studies, which show that financially literate individuals are less likely to delegate financial decisions (Calcagno and Monticone, 2015) and less likely to follow financial advice once it is provided (Stolper, 2018; Hackethal et al., 2010). According to this logic, efficient portfolio diversification can be achieved either through financial literacy or financial advice (Gaudecker, 2015). However, our findings suggest that financial education enhances diversification while increasing advice taking. As a result, financial education might increase the probability of achieving favorable investment outcomes, both directly (through enhanced diversification ability) and indirectly (through increased delegation, assuming that delegation improves investment outcomes). Thus, our study makes a more optimistic case for the effect of financial education interventions on financial outcomes, at least in the short run, than previous studies (cf. Fernandes et al., 2014).

Building on the findings documented in this study, several interesting extensions are worth pointing out. First, we cannot clearly distinguish whether the financial education intervention increased delegation because of the improvement in financial knowledge or because of increased transparency regarding the decision-making process. To test whether transparency per se affects the delegation decision, our design could be extended to investigate whether the

treatment effect differs when participants can delegate decision-making to a human asset manager. As transparency might be more important for an algorithmic than for a human decision-maker (Dietvorst et al., 2015), a stronger increase in delegation to the robo-advisor would indicate that transparency per se affects the delegation decision.

Second, the spillover effects of the financial education intervention cannot be disentangled from the delegation decision in our framework since participants simultaneously optimize over the self-managed and delegated portfolios. Thus, the effect of the intervention on investment behavior could be studied with and without the availability of a robo-advisory tool. This design could also be used to test whether the treatment primarily affected the participants' perception of the decision-making task or the robo-advisor's service.

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